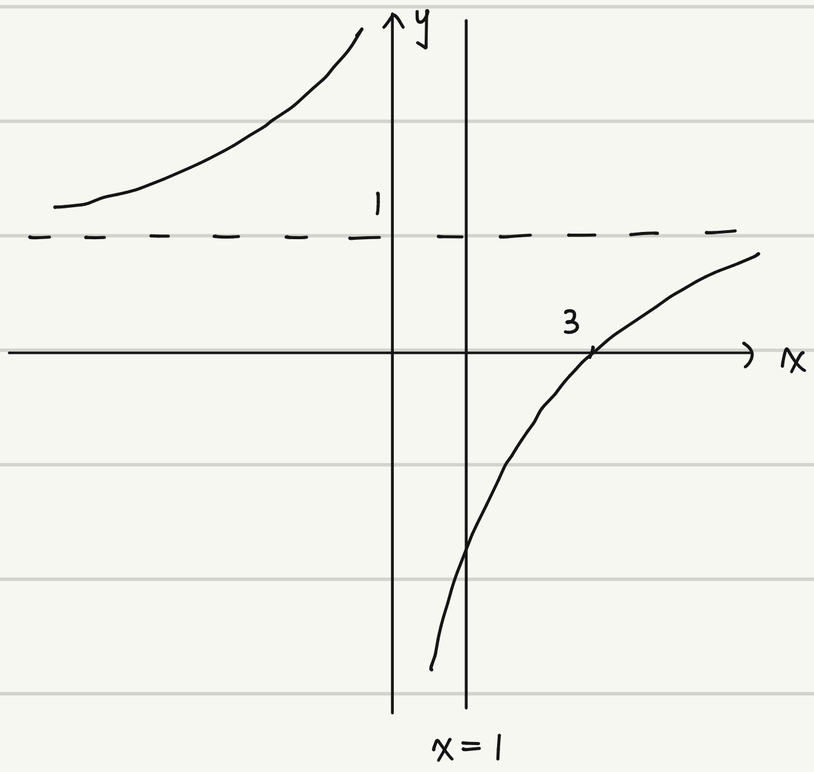
9. 2

2 (1), # 4. # 5

2

(1) $y = 1 - \frac{3}{x}$, $y = 0$, $x = 1$



$$\pi \int_1^3 \left(1 - \frac{3}{x}\right)^2 dx$$

$$= \pi \int_1^3 \left(1 - \frac{6}{x} + \frac{9}{x^2}\right) dx$$

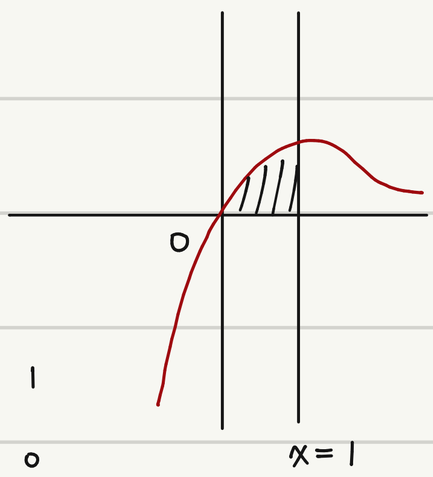
$$= \pi \left[x - 6 \ln x - \frac{9}{x} \right]_1^3$$

$$= \pi \left(3 - 6 \ln 3 - 3 - 1 + 9 \right)$$

$$= \pi \left(8 - 6 \ln 3 \right) \text{ "}$$

4

(1)



$$\int_0^1 x \cdot e^{-x} \, dx$$

$$= \left[x \cdot (-e^{-x}) - e^{-x} \right]_0^1$$

$$= \left[-e^{-x} (x+1) \right]_0^1$$

$$= -2e^{-1} + 1$$

$$= 1 - \frac{2}{e} \text{ "}$$

(2) $$\pi \int_0^1 x^2 \cdot e^{-2x} \, dx$$

$$= \pi \left[x^2 \cdot \left(-\frac{1}{2}\right) e^{-2x} - 2x \cdot \frac{1}{4} e^{-2x} + 2 \cdot \left(-\frac{1}{8}\right) e^{-2x} \right]_0^1$$

$$= \pi \left(-\frac{1}{2} e^{-2} - \cancel{2} \cdot \frac{1}{\cancel{4}_2} e^{-2} - \frac{1}{4} e^{-2} + \frac{1}{4} \right)$$

$$= \pi \left(-\frac{5}{4} e^{-2} + \frac{1}{4} \right) \text{ "}$$

5

$$\int_1^e \ln x \, dx = 1$$

$$\pi \int_1^e (\ln x)^2 \, dx$$

$$= \pi \left(\left[(\ln x)^2 \cdot x \right]_1^e - \int_1^e 2 \ln x \cdot \frac{1}{x} \cdot x \, dx \right)$$

$$= \pi \left(e - 2 \right) \quad //$$